Babak Asadollah

Seyed Babak Haji Seyed Asadollah
Nationality: Iranian

EDUCATION

Ph.D. in Environmental Resource Engineering *08/2022 – Expected 12/2026*
Concentration: Geospatial Information Science & Engineering
State University of New York (SUNY), College of Environmental Science and Forestry (ESF)
Supervised by Prof. Giorgos Mountrakis, Full Professor. Research focus: satellite-based heat hazard modeling and environmental justice/health impact assessment.

M.Eng. in Civil Engineering *09/2014 – 09/2017*
Water Engineering & Hydraulic Structures — Islamic Azad University, South Tehran Branch
Supervised by Dr. Hooman Hajikandi, Assistant Professor. Thesis: "Numerical study of hydraulic characteristics of a submerged jet entering the dense ambient".

B.Eng. in Civil Engineering *09/2010 – 09/2014*
Islamic Azad University, Central Tehran Branch

RESEARCH SUMMARY

Babak Asadollah's research uses remote sensing and machine learning for local-scale weather and hazard forecasting with uncertainty quantification — including extreme heat, precipitation, and drought — integrating satellite data with numerical model outputs and ground observations to assess environmental, agricultural, and public health impacts. His work spans over 30 peer-reviewed journal articles appearing in journals including the ISPRS Journal of Photogrammetry and Remote Sensing and Water Research, several conference papers (including two oral presentations at the American Geophysical Union (AGU) Fall Meeting) and a book chapter. According to Google Scholar, his publications have been cited more than 1,490 times since 2020, with a current H-index of 16. His work applying machine learning to water quality estimation has been cited over 450 times to date — well above the field average of 9.02 citations for articles published in the Engineering category in 2021. He maintains active research collaborations with scholars in Spain, the United Kingdom, Malaysia, and the Middle East, and has reviewed more than 50 manuscripts for journals including Science of The Total Environment, the Journal of Hydrology, the International Journal of Applied Earth Observation and Geoinformation, and Environmental Science and Ecotechnology.

RESEARCH EXPERIENCE

Research Assistant — NYS Center of Excellence in Healthy Water Solutions project
Supervisor: Giorgos Mountrakis
- Supported by an award from the NYS Center of Excellence in Healthy Water Solutions (SUNY ESF / Clarkson University; NYSTAR-funded), secured by Prof. Mountrakis on the basis of the SHINE methodology (Haji Seyed Asadollah et al., ISPRS Journal of Photogrammetry and Remote Sensing, 2026).

Research Assistant — Developing Satellite-based heat Index estimatioN modEl (SHINE) methodology
Supervisor: Giorgos Mountrakis | 06/2024 – 01/2025
- Compensated via a SUNY Research Foundation fellowship in lieu of a standard assistantship

Research Assistant — Developing novel deep learning classifiers for remote sensing imagery
Supervisor: Giorgos Mountrakis | 06/2023 – 01/2024
- Compensated via a SUNY Research Foundation

Research Assistant — Development of a flood forecasting system coupling meteorological, hydrologic, and hydraulic models
Supervisor: Ahmad Sharafati | 09/2018 – 07/2022
Islamic Azad University Science and Research Branch

TEACHING EXPERIENCE

Teaching Assistant, SUNY ESF

- Spring 2025 & 2026 — ERE 565, Principles of Remote Sensing (4 credits)
- Spring 2024 — ERE 133, Intro to Engineering Design (3 credits)
- Spring 2023 — ERE 565, Principles of Remote Sensing (4 credits)
- Fall 2022 — GNE 271, Statics (3 credits)

Instructor, Open Academy Teaching & Learning Center (TLC), SUNY ESF

Summer 2025

- Trained incoming graduate teaching, research, and graduate assistants on the responsibilities and best practices of GA/TA/RA roles.

Lecturer, Experimental Physics Lab, SUNY ESF

Fall 2025

- Sole instructor of record for the laboratory (experimental) section of the introductory physics course (EPH 201 and 202).

Lecturer, Islamic Azad University

Under the supervision of Dr. Ahmad Sharafati between years 2019 to 2021

- Soft Computing (CE4001)
- Flood Engineering and Management (CE4623)
- Advanced Engineering Hydrology (CE4701)

TEACHING CERTIFICATIONS & PEDAGOGICAL TRAINING

Certificate in University Teaching (CUT), Future Professoriate Program (FPP)

Fall 2026 – Spring 2027 (in progress)

- Completing an eight-part seminar series on instructional design, active learning, lesson planning, and assessment design, together with departmental teaching requirements and faculty-liaison mentorship.

PUBLICATIONS

Journal Articles (33 listed)

33. SMART-Soil: Satellite-based machine learning approach for reliable tracking of soil moisture across CONUS (2026)
Seyed Babak Haji Seyed Asadollah, Jodar-Abellan, A., & Pardo, M. Á.
Applied Soil Ecology | https://doi.org/10.1016/j.apsoil.2026.107201

32. Satellite-based heat Index estimatioN modEl (SHINE): An integrated machine learning approach for the conterminous United States (2026)
Seyed Babak Haji Seyed Asadollah, Mountrakis, G., & Shaw, S. B.
ISPRS Journal of Photogrammetry and Remote Sensing | https://doi.org/10.1016/j.isprsjprs.2026.01.018

31. Climate-responsive crop forecasting: an EEMD-LSTM fusion approach for improved strategic crop yield simulation (2026)
Seyed Babak Haji Seyed Asadollah, Kheyruri, Y., Sharafati, A., & Hameed, A. S.
Acta Geophysica | https://doi.org/10.1007/s11600-025-01764-6

30. A hybrid EMD-DFA–LSTM–MOOTLBO model for accurate water quality index prediction (2025)
Ariyaei, A., Kheyruri, Y., Doroudi, S., Sharafati, A., Hameed, A. S., Ghozat, A., & Seyed Babak Haji Seyed Asadollah
Journal of Water Process Engineering | https://doi.org/10.1016/j.jwpe.2025.108558

29. Enhanced monocular depth estimation using novel scale-invariant Error Structure Similarity Index measure optimization in Convolutional Neural network architecture (2025)
Hemmati, E., Jarahizadeh, S., Aghabalaei, A., & Seyed Babak Haji Seyed Asadollah
Journal of Visual Communication and Image Representation | https://doi.org/10.1016/j.jvcir.2025.104531

28. Evaluating water quality using the Geographical Detector Model in Hong Kong (2025)
Kheyruri, Y., Sharafati, A., Seyed Babak Haji Seyed Asadollah, Ariyaei, A., Hameed, A. S., & Farzad, R.
Physics and Chemistry of the Earth, Parts A/B/C | https://doi.org/10.1016/j.pce.2025.103971

27. Dissolved organic carbon estimation in lakes: Improving machine learning with data augmentation on fusion of multi-sensor remote sensing observations (2025)
Seyed Babak Haji Seyed Asadollah, Safaeinia A., Jarahizadeh S., Alcala F. J., Sharafati A., Jodar-Abellan, A.
Water Research (Impact score = 12.4) | https://doi.org/10.1016/j.watres.2025.123350

26. Exploring expert perceptions towards emerging pollutants and their impacts in reused wastewater and agriculture (2024)
Jodar-Abellan, A., Albaladejo-García, J. A., Aznar-Crespo, P., de los Santos, M. B., Seyed Babak Haji Seyed Asadollah, Derdour, A., et al., & Serrano-Bernardo, F.
Agricultural Water Management | https://doi.org/10.1016/j.agwat.2024.109098

25. Assessing current and future available resources to supply urban water demands using a high-resolution SWAT model coupled with recurrent neural networks and validated through the SIMPA model in karstic Mediterranean environments (2024)
Jodar-Abellan, A., Pardo, M. Á., Seyed Babak Haji Seyed Asadollah, & Bailey, R. T.
Environmental Science and Pollution Research | https://doi.org/10.1007/s11356-024-34404-5

24. Optimizing machine learning for agricultural productivity: A novel approach with RScv and remote sensing data over Europe (2024)
Seyed Babak Haji Seyed Asadollah, Jodar-Abellan, A., & Pardo, M. Á.
Agricultural Systems | https://doi.org/10.1016/j.agsy.2024.103955

23. Examining the effects of meteorological drought variability on rainfed cultivation yields in Iran (2024)
Kheyruri, Y., Seyed Babak Haji Seyed Asadollah, Sharafati, A., Rahman, A., & Mohajeri, S. H.
Theoretical and Applied Climatology | https://doi.org/10.1007/s00704-024-05013-6

22. Satellite-based ensemble intelligent approach for predicting forest fire: a case of the Hyrcanian forest in Iran (2024)
Seyed Babak Haji Seyed Asadollah, Sharafati, A., & Motta, D.
Environmental Science and Pollution Research | https://doi.org/10.1007/s11356-024-32615-4

21. Long-Term Water Quality Monitoring: Using Satellite Images for Temporal and Spatial Monitoring of Thermal Pollution in Water Resources (2024)
Naimaee, R., Kiani, A., Jarahizadeh, S., Seyed Babak Haji Seyed Asadollah, Melgarejo, P., & Jodar-Abellan, A.
Sustainability | https://doi.org/10.3390/su16020646

20. Estimation of soil moisture from remote sensing products using an ensemble machine learning model: a case study of Lake Urmia Basin, Iran (2023)
Seyed Babak Haji Seyed Asadollah, Sharafati, A., Saeedi, M., & Shahid, S.
Earth Science Informatics | https://doi.org/10.1007/s12145-023-01172-8

19. A novel intelligent approach for predicting meteorological drought based on satellite-based precipitation product: Application of an EMD-DFA-DBN hybrid model (2023)
Ghozat, A., Sharafati, A., Seyed Babak Haji Seyed Asadollah, & Motta, D.
Computers and Electronics in Agriculture | https://doi.org/10.1016/j.compag.2023.107946

18. Satellite-based prediction of surface dust mass concentration in southeastern Iran using an intelligent approach (2023)
Seyed Babak Haji Seyed Asadollah, Sharafati, A., Motta, D., Jodar-Abellan, A., & Pardo, M. Á.
Stochastic Environmental Research and Risk Assessment | https://doi.org/10.1007/s00477-023-02473-6

17. Developing a generic relation for predicting sediment pick-up rate using symbolic soft computing techniques (2023)
Haghbin, M., Sharafati, A., Seyed Babak Haji Seyed Asadollah, & Motta, D.
Environmental Science and Pollution Research | https://doi.org/10.1007/s11356-022-23450-6

16. An intelligent approach for estimating aeration efficiency in stepped cascades: optimized support vector regression models and mutual information theory (2022)
Seyed Babak Haji Seyed Asadollah, Sharafati, A., Haghbin, M., Motta, D., & Hosseinian Moghadam Noghani, M.
Soft Computing | https://doi.org/10.1007/s00500-022-07437-6

15. Post-processing of the UKMO ensemble precipitation product over various regions of Iran: integration of long short-term memory model with principal component analysis (2022)
Alizadeh, S., Seyed Babak Haji Seyed Asadollah, & Sharafati, A.
Theoretical and Applied Climatology | https://doi.org/10.1007/s00704-022-04170-w

14. Determination of discharge coefficient of stepped morning glory spillway using a hybrid data-driven method (2022)
Haghbin, M., Sharafati, A., Aghamajidi, R., Seyed Babak Haji Seyed Asadollah, Hosseinian Moghadam Noghani, M., & Jalón, M. L.
Flow Measurement and Instrumentation | https://doi.org/10.1016/j.flowmeasinst.2022.102161

13. A robust stochastic approach in correcting the TRMM precipitation product and simulating flood features (2022)
Seyed Babak Haji Seyed Asadollah, Sharafati, A., Neshat, A., & Hemmati, N.
Environmental Monitoring and Assessment | https://doi.org/10.1007/s10661-022-10030-x

12. Application of ensemble machine learning model in downscaling and projecting climate variables over different climate regions in Iran (2022)
Seyed Babak Haji Seyed Asadollah, Sharafati, A., & Shahid, S.
Environmental Science and Pollution Research | https://doi.org/10.1007/s11356-021-16964-y

11. An efficient strategy for predicting river dissolved oxygen concentration: application of deep recurrent neural network model (2021)
Valizadeh Moghadam, S., Sharafati, A., Feizi, H., Marjaie, S. M. S., Seyed Babak Haji Seyed Asadollah, & Motta, D.
Environmental Monitoring and Assessment | https://doi.org/10.1007/s10661-021-09586-x

10. Application of bagging ensemble model for predicting compressive strength of hollow concrete masonry prism (2021)
Sharafati, A., Seyed Babak Haji Seyed Asadollah, & Al-Ansari, N.
Ain Shams Engineering Journal | https://doi.org/10.1016/j.asej.2021.03.028

9. Prediction of heat waves using meteorological variables in diverse regions of Iran with advanced machine learning models (2021)
Seyed Babak Haji Seyed Asadollah, Khan, N., Sharafati, A., Shahid, S., Chung, E.-S., & Wang, X.-J.
Stochastic Environmental Research and Risk Assessment | https://doi.org/10.1007/s00477-021-02103-z

8. Estimating the transient storage parameters for pollution modeling in small streams: a comparison of newly developed hybrid optimization algorithms (2021)
Ehteram, M., Sharafati, A., Seyed Babak Haji Seyed Asadollah, & Neshat, A.
Environmental Monitoring and Assessment | https://doi.org/10.1007/s10661-021-09269-7

7. Assessing the impact of climate change on urban water demand and related uncertainties: a case study of Neyshabur, Iran (2021)
Sharafati, A., Seyed Babak Haji Seyed Asadollah, & Shahbazi, A.
Theoretical and Applied Climatology | https://doi.org/10.1007/s00704-021-03638-5

6. River Water Quality Index prediction and uncertainty analysis: a comparative study of machine learning models (2021)
Seyed Babak Haji Seyed Asadollah, Sharafati, A., Motta, D., & Yaseen, Z. M.
Journal of Environmental Chemical Engineering | https://doi.org/10.1016/j.jece.2020.104599

5. Machine learning model development for predicting aeration efficiency through Parshall flume (2021)
Sangeeta, Seyed Babak Haji Seyed Asadollah, Sharafati, A., Sihag, P., Al-Ansari, N., & Chau, K.-W.
Engineering Applications of Computational Fluid Mechanics | https://doi.org/10.1080/19942060.2021.1922314

4. A new artificial intelligence strategy for predicting the groundwater level over the Rafsanjan aquifer in Iran (2020)
Sharafati, A., Seyed Babak Haji Seyed Asadollah, & Neshat, A.
Journal of Hydrology | https://doi.org/10.1016/j.jhydrol.2020.125468

3. Application of newly developed ensemble machine learning models for daily suspended sediment load prediction and related uncertainty analysis (2020)
Sharafati, A., Seyed Babak Haji Seyed Asadollah, Motta, D., & Yaseen, Z. M.
Hydrological Sciences Journal | https://doi.org/10.1080/02626667.2020.1786571

2. The potential of new ensemble machine learning models for effluent quality parameters prediction and related uncertainty (2020)
Sharafati, A., Seyed Babak Haji Seyed Asadollah, & Hosseinzadeh, M.
Process Safety and Environmental Protection | https://doi.org/10.1016/j.psep.2020.04.045

1. Scouring Depth Assessment Downstream of Weirs Using Hybrid Intelligence Models (2020)
Sharafati, A., Haghbin, M., Seyed Babak Haji Seyed Asadollah, Tiwari, N. K., Al-Ansari, N., & Yaseen, Z. M.
Applied Sciences | https://doi.org/10.3390/app10113714

Conference Papers (6 listed)

6. Predicción del rendimiento de los cuatro tipos principales de cultivos europeos mediante variables climático-ambientales obtenidas por teledetección: resultados preliminares (2026)
(Prediction of yield for the four main European crop types using climate-environmental variables derived from remote sensing: preliminary results)
Seyed Babak Haji Seyed Asadollah, Pardo, M. Á., & Jódar-Abellán, A.
8th National Water Congress: "Food Security," 11–12 September 2026, Centro de Congresos de Elche, Alicante, Spain.

5. Real-Time Flood Risk Prediction Using Temporal Transformer-based Deep Learning and Multisource Remote Sensing Data (2025)
Seyed Babak Haji Seyed Asadollah, Safaeinia, A., Jarahizadeh, S., & Jarahizadeh, S.
American Geophysical Union (AGU), December 15–19, 2025, New Orleans, LA — Oral Presentation. Session: "Recent Advances in Flood Risk: Prediction, Monitoring, Assessment, Management, Mitigation, and Adaptation Planning I," 18 December 2025, 10:30–12:00 CST.

4. Pinpointing High-Yield Climate Windows via a Satellite-Powered RF-GDM Framework for Four Staple Crops (2025)
Seyed Babak Haji Seyed Asadollah, Jodar-Abellan, A., & Picazo, M. Á. P.
American Geophysical Union (AGU), December 15–19, 2025, New Orleans, LA — Oral Presentation. Session: "Data-Driven Agriculture: Remote Sensing and Machine Learning Solutions for Food Security II," 16 December 2025, 14:15–15:45 CST.

3. Evaluación de los recursos hídricos convencionales destinados al abastecimiento urbano en una cuenca hidrológica agro-forestal del sureste de España (2024)
(Evaluation of conventional water resources for urban supply in an agro-forestry watershed in southeastern Spain)
Jodar-Abellan, A., Van Oudenhove, M., Fernández-Soler, C., & Seyed Babak Haji Seyed Asadollah
6th National Water Congress: "Water, Agriculture and Food," 7–8 October 2024, Alicante, Spain.

2. Determinación del umbral de escorrentía y disponibilidad hídrica de la cuenca hidrográfica del río Jubones, Ecuador (2023)
(Determination of the runoff threshold and water availability in the Jubones River watershed, Ecuador)
Palacios Cabrera, T. A., Jodar-Abellan, A., Gómez, D. N., Melgarejo, P., Abdessamed, D., Bailey, R., & Seyed Babak Haji Seyed Asadollah
5th National Water Congress: "Water Security," 14–15 September 2023, Alicante, Spain.

1. Assessing urban water supply from karstic groundwater reservoirs through two hydrological models and the Exploitation Index in the southeast of Spain (2023)
Brazales Cervantes, P., Seyed Babak Haji Seyed Asadollah, & Jódar Abellán, A.
European Geosciences Union (EGU), 23–28 April 2023, Vienna, Austria.

Book Chapters

A Case Study of Evaluating Agricultural and Forest Land Conversion into Urban Areas and the Related Increase of Peak Runoff Rates with GIS-Based Tools and the SWAT Model: Southeast Spain (2023)
Jodar-Abellan, A., Palacios-Cabrera, T., Haji Seyed Asadollah, S. B., Legua, P., Abdessamed, D., & Melgarejo, P.
In: Teaching in Land Management and Applied Geospatial Tools: The Use of Geographic Information Systems and Remote Sensing Data to Design Advanced Technical Interventions for Sustainable Land Management, Ch. 3, pp. 53–80. Nova Science Publishers, Hauppauge, NY. ISBN 979-8-89113-194-1.
https://doi.org/10.52305/QUKP6045

FUNDING & FELLOWSHIPS

- **Summer Research Fellowship**, Dept. of Environmental Resources Engineering, SUNY ESF (Chair: Dr. Stephen Shaw) — $4,000 (Summer 2026)
- **Summer Research Fellowship**, Dept. of Environmental Resources Engineering, SUNY ESF (Chair: Dr. Stephen Shaw) — $3,000 (Summer 2025)
- **On-Campus Internship Incremental Funding**, Office of Career Services, SUNY ESF — $2,000 (Summer 2025)

(See also the SUNY Research Foundation fellowship supporting the Mountrakis-supervised RA position, listed under Research Experience.)

Grant Proposals (Submitted)

Development of a Nationwide Heat Index Estimation Model for Environmental Justice and Health Impacts Assessment
NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) — Solicitation NNH24ZDA001N
Role: Future Investigator (FI); Advisor: Giorgos Mountrakis
Status: Submitted, ROSES 2024 cycle — not selected

AWARDS & HONORS

- **GSA Travel Grant**, ESF Graduate Student Association — $500 (2025–2026 award year)

PROFESSIONAL DEVELOPMENT

- **SUNY Startup Summer School**, Certificate of Completion — SUNY & SUNY Research Foundation (July 2026)

ACADEMIC & PROFESSIONAL SERVICE

Chairperson, Graduate Research & Curriculum Committee — Graduate Student Association (GSA), SUNY ESF
Fall 2023 – Spring 2024; Fall 2024 – Spring 2025

- Chaired the GSA Research & Curriculum Committee and served as GSA's representative to the Academic Governance Committee on Research (COR) and Academic Affairs Committees.
- Led annual planning for the CNY Conference on Environmental Science and Studies.
- Worked with the Vice President for Research, Provost, and Department Chairs to evaluate and improve conditions for graduate research and learning — including funding packages, lab space and equipment, safety training, grant awareness, and endowed fellowships — and proposed improvements.
- Compiled graduate research productivity metrics, benchmarked performance against peer programs outside ESF, and contributed material highlighting student achievement for donor and alumni engagement.
- Reviewed ESF graduate courses with the Dean of the Graduate School and the Curriculum Committee, using course evaluations and other assessment data to identify concerns and propose upgrades.
- Prepared and presented summaries of Committee activities to the ESF Board of Trustees upon request from the GSA President and Board of Trustees Representative.

PROFESSIONAL MEMBERSHIPS

- Institute of Electrical and Electronics Engineers, Geoscience & Remote Sensing Society (IEEE GRSS) — 2026
- American Geophysical Union (AGU) — 2025 and 2026

EDITORIAL SERVICE

Guest Editor: Journal of Physics and Chemistry of the Earth, Parts A/B/C – Starting June 2026

- Special Issue title: Intelligent Hydro-Meteorological Systems: From Data-Driven Forecasting to Early Warning for Extreme Events

PEER REVIEW SERVICE

Ad hoc reviewer for more than 50 manuscripts for the following journals:

Elsevier: Applied Mathematical Modelling; Science of The Total Environment; Journal of Environmental Chemical Engineering; Agricultural and Forest Meteorology; Agricultural Systems; Environmental Modeling and Software; Environmental Science and Ecotechnology; Intelligent Systems with Applications; International Journal of Applied Earth Observation and Geoinformation; Journal of Hydrology; Remote Sensing Applications: Society and Environment

Springer: Applied Water Science; Climatic Change; Earth Science Informatics; Environmental Monitoring and Assessment; Environmental Sciences Europe; KSCE Journal of Civil Engineering; Modeling Earth Systems and Environment; Stochastic Environmental Research and Risk Assessment; Theoretical and Applied Climatology

MDPI: Remote Sensing; Toxics; Water

Nature Portfolio: Scientific Reports

TECHNICAL SKILLS

- **Programming Languages:** Python (Advanced), R (Intermediate), MATLAB (Intermediate), Julia (Beginner)
- **ML/DL Frameworks:** TensorFlow (Advanced), PyTorch (Intermediate), scikit-learn (Advanced), Keras (Advanced)
- **Remote Sensing & GIS:** ArcGIS Pro (Advanced, I was the instructor for the ERE565 course), Google Earth Engine (Intermediate), Pix4D (Intermediate), Geomatica (Intermediate)
- **Hydrologic & Water Modeling Software:** HEC-RAS (Intermediate), SWAT (Intermediate), EPANET (Beginner), MIKE (Beginner)
- **Computing:** Git/GitHub; HPC/cluster computing on Syracuse University's Orange Grid (Linux environment)
- **Spoken Languages:** Farsi (Native), English (Fluent), Spanish (Beginner)

REFERENCES

- **Prof. Giorgos Mountrakis**
  Full Professor, Department of Environmental Resources Engineering
  State University of New York College of Environmental Science and Forestry (SUNY ESF)
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  Work Phone: (315) 470-4824
  Website: www.aboutgis.com
  Relationship: Ph.D. Advisor

- **Prof. Miguel Ángel Pardo Picazo**
  Full Professor, Department of Civil Engineering
  University of Alicante (UA)
  Email: mpardo@ua.es
  Relationship: Former Supervisor

- **Dr. Ahmad Sharafati**
  Associate Professor, Department of Civil Engineering
  Islamic Azad University, Science and Research Branch, Tehran
  Email: asharafati@iau.ac.ir
  Relationship: Former Research Supervisor